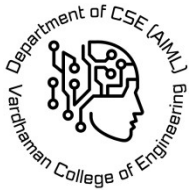


VARDHAMAN COLLEGE OF ENGINEERING, HYDERABAD



Autonomous institute affiliated to JNTUH
DEPARTMENT OF CSE(AI&ML)
IV-Year Major Project Abstract Level Review



Title: Vision Language Robotics for Collaborative Manufacturing

Domain: Artificial Intelligence | Computer Vision | Robotics | Industry 4.0

BATCH ID : 03

S. No	Roll. No	Student Name
1	23881A6623	Hanumakonda Aryan
2	23881A6639	Kottam Eshwar Reddy
3	23881A6654	Sanaboina Rathik Raj

Supervisor

Dr. Ramachandro Majji

Associate Professor

Department of Computer Science And Engineering (AI & ML)

Outlines

- **Abstract**
- **Introduction**
- **Literature Review**
- **Existing System with Pros and Cons (comparison with existing systems- Table)**
- **Objectives & Expected Deliverables**
- **SDG Mapping**
- **PO Mapping**
- **PSO Mapping**
- **Timeline**
- **References**

Abstract

- Collaborative robots are expected to work alongside human operators in dynamic production environments where the assembly task itself can change mid-shift.
- Current Vision-Language-Action approaches generalize well across objects and instructions but operate under a fixed-task assumption, leaving real-time task adaptation unaddressed.
- This work proposes a framework that lets a collaborative robot reinterpret its own task plan when the operator issues a changeover instruction, rather than requiring manual reconfiguration.
- The system builds on OpenVLA as the manipulation backbone, adding a re-planning layer that converts natural-language changeover commands into an updated subtask sequence.
- Scene understanding and variant identification are handled through vision-language grounding, keeping the robot aware of which product configuration is currently active.
- The framework will be evaluated in simulation against existing VLM-based baselines, measuring task success and how quickly the system adapts once a changeover is issued.

Introduction

- Manufacturing is moving toward greater flexibility, with human operators and robots increasingly expected to work side by side rather than in separated, fixed roles.
- This shift is part of the broader Industry 5.0 movement, which prioritizes adaptability and human-centered production over rigid automation.
- Advances in large language models and vision-language models have opened a new path for robots to understand natural instructions and visual context together, rather than relying on hard-coded rules.
- This has given rise to Vision-Language-Action models, which connect perception, language understanding, and physical action in a single pipeline.
- Early results in this space have been promising, with robots showing an ability to generalize across new objects, environments, and phrased instructions without task-specific reprogramming.
- Our project builds on this momentum, applying it specifically to the unresolved problem of task adaptation in collaborative manufacturing settings.

Existing System

❖ **Traditional industrial robots**

Pre-programmed, fixed-sequence robots with no language or visual understanding , reprogrammed manually for every task change.

❖ **General-purpose VLA models (RT-2, OpenVLA)**

Use vision-language pretraining to generalize manipulation across objects and instructions, but assume one fixed task.

❖ **VLM-guided HRC systems (Fan & Zheng, 2024)**

Combine CLIP + LLM to resolve ambiguous human instructions (e.g., "hand me that one") within a fixed assembly task.

❖ **Common limitation**

All existing systems assume the task itself stays constant — none adapt the plan when the product or task changes mid-shift.

Pros and Cons of Existing System

Pros:

- Strong generalization to new objects and language instructions (RT-2, OpenVLA)
- Open-source and fine-tunable on custom data (OpenVLA)
- Resolves referential ambiguity in human commands (Fan & Zheng, 93.3% success)
- High success rates reported in controlled benchmark settings

Cons:

- Assumes a fixed, unchanging assembly task
- No mechanism to adapt the task plan in real time
- Still requires manual reprogramming when the product/variant changes
- Precision insufficient for fine industrial assembly tasks
- High computational requirements, not optimized for manufacturing deployment

Problem Statement

- ❖ **High-mix manufacturing changes products frequently.**

Industries like aerospace, medical devices, and custom electronics switch product variants multiple times per shift.

- ❖ **Robots can't keep up without manual reprogramming.**

Every variant switch forces a stop-and-reprogram cycle, causing downtime and reducing flexibility.

- ❖ **Existing VLA models don't solve this.**

RT-2, OpenVLA, and even ambiguity-resolution work like Fan & Zheng (2024) assume a fixed task — none adapt the plan itself when the task changes.

- ❖ **No system currently lets a robot adapt its plan in real time from a spoken instruction.**

This is the exact gap our project addresses.

Ref	Paper	Year	Methodology	Dataset	Performance Metrics	Key Findings	Research Gap
[1]	Vision-Language Model-Based Human-Robot Collaboration for Smart Manufacturing: A State-of-the-Art Survey	2025	Systematic survey of VLMs in collaborative manufacturing	109 selected research papers	Reviews 109 publications covering assembly, inspection and HRC	Provides taxonomy, industrial challenges and future research directions	No implementation or quantitative benchmarking
[2]	Vision-Language-Action Models for Robotics: A Review Towards Real-World Applications	2025	Comprehensive survey of VLA architectures, datasets, benchmarks and deployment strategies	Survey Literature	Reviews 100+ VLA studies and compares evaluation protocols	Summarizes recent advances, benchmarks and deployment challenges	No experimental implementation or quantitative evaluation
[3]	Vision-Language Models for Human-Robot Collaboration	2025	Review of multimodal perception, language understanding and collaborative robotics	Human-Robot Collaboration Literature	Comparative analysis of recent VLM-based HRC frameworks	Summarizes multimodal reasoning and collaborative task execution	Limited discussion on manufacturing deployment and industrial validation
[4]	A vision-language-guided robotic action planning approach for ambiguity mitigation in HRC manufacturing (Fan & Zheng)	2024	CLIP for visual grounding + LLM-based referred-object retrieval + action planner	Custom industrial HRC assembly dataset	93.3% success rate	Resolves ambiguous object references ("hand me that one") in a fixed assembly task	No mechanism to adapt the task/plan itself when the manufacturing goal changes (e.g., variant switch)
[5]	OpenVLA: An Open-Source Vision-Language-Action Model	2024	Open-source Vision-Language-Action (VLA) model for robotic manipulation	Open X-Embodiment	16.5% absolute improvement over RT-2-X across 29 tasks (verify exact base success-rate figure against the paper before citing)	Achieves state-of-the-art manipulation performance with efficient fine-tuning and open-source deployment	Requires high computational resources and lacks manufacturing-specific optimization
[6]	QUAR-VLA: Vision-Language-Action Model for Quadruped Robots	2024	Vision-Language-Action framework for autonomous navigation and manipulation on quadruped robots	QUARD (259K simulated + 3K real-world episodes)	Trained on 262K demonstrations with strong sim-to-real generalization	Enables language-guided locomotion, manipulation and reasoning	Focused on quadruped robots instead of collaborative industrial manipulators
[7]	Towards Open-World Grasping with Large Vision-Language Models	2024	VLM integrated with segmentation, grasp synthesis and contact reasoning	OCID-VLG, Grasp-MultiObject, Real Robot Evaluation	Outperforms previous supervised and zero-shot LLM baselines in simulation and real hardware	Improves zero-shot grasp planning for unseen objects	Limited to grasping rather than complete manufacturing workflows
[8]	RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control	2023	Co-fine-tunes a VLM on web-scale + robot trajectory data, outputs	Internet-scale image-text + robot demonstration data	62% success rate (custom evaluation)	First major VLA; shows web-scale semantic knowledge transfers to	General manipulation only; no manufacturing-

Objectives & Deliverables Proposed System

Objective 1: Develop a Vision-Language Robotics framework capable of understanding natural language instructions and visual inputs for collaborative manufacturing tasks.

Objective 2: Implement multimodal learning techniques that combine computer vision, language models, and robotic control for intelligent task planning and execution.

Objective 3: Design a collaborative robotic system that improves manufacturing flexibility, task accuracy, human-robot interaction, and workplace safety.

Expected Deliverables:

1. An intelligent Vision-Language robotic prototype capable of understanding natural language instructions and executing collaborative manufacturing tasks.
2. A real-time multimodal robotic system integrating computer vision, language reasoning, and autonomous task planning for smart manufacturing.

SDG Mapping

SDG No.	Goal Title	Justification
SDG 4	Quality Education	The project promotes learning and research in artificial intelligence, robotics, computer vision, and intelligent automation by providing a practical implementation of Vision-Language Robotics for engineering education and skill development.
SDG 8	Decent Work and Economic Growth	By automating repetitive manufacturing tasks and enabling safe human-robot collaboration, the system enhances workforce productivity, improves operational efficiency, and supports sustainable industrial growth.
SDG 9	Industry, Innovation and Infrastructure	The proposed framework integrates Vision-Language Models, computer vision, and collaborative robotics to develop intelligent manufacturing solutions, fostering industrial innovation and smart factory infrastructure.
SDG 12	Responsible Consumption and Production	Production The robotic system improves production efficiency, minimizes material waste, reduces manufacturing errors, and optimizes resource utilization through intelligent decision-making and automation.
SDG 17	Partnerships for the Goals	the Goals The project encourages collaboration among academia, industry, and AI research communities by integrating open-source robotics frameworks, Vision-Language Models, and advanced manufacturing technologies.

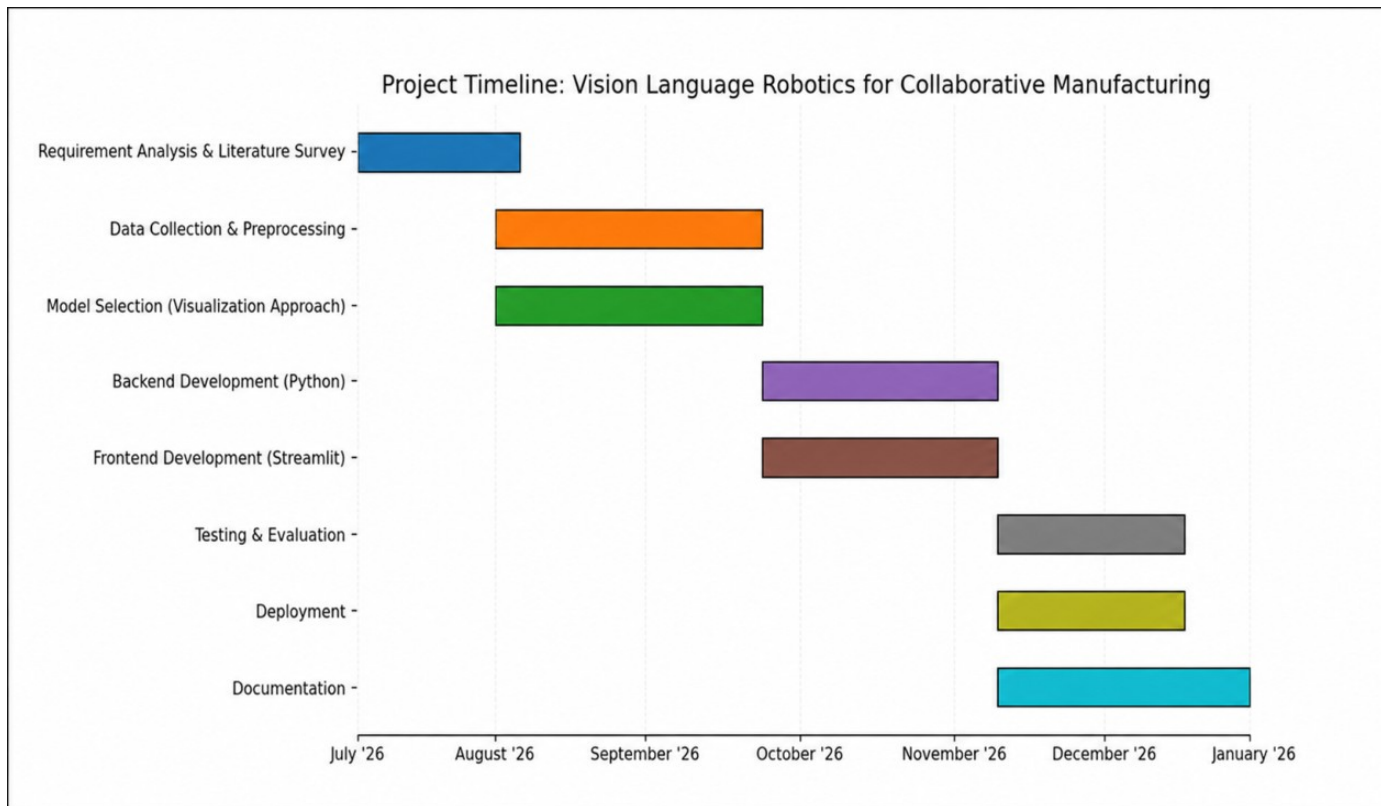
PO Mapping

Po number	Name of the PO	Targeted Justification
PO1	Engineering knowledge	Apply principles of Artificial Intelligence, Computer Vision, Machine Learning, Robotics for collaborative manufacturing.
PO2	Problem analysis	Analyze challenges in conventional industrial automation, to design effective AI-driven manufacturing solutions.
PO3	Design/Development of Solutions	Design and develop a multimodal robotic framework integrating Vision-Language Models, computer vision, and execution in manufacturing environments.
PO5	Engineering Tool Usage	Utilize modern engineering tools and technologies implement and evaluate the proposed system.
PO7	Ethics	Develop an AI-enabled robotic system that emphasizes safe, reliable, and ethical human-robot collaboration.
PO8	Individual and Collaborative Team Work	Work effectively as an individual and as a team to design, develop, integrate, test, and validate the Vision-Language robotic system through collaborative project execution
PO11	Life-Long Learning	Continuously explore emerging to develop innovative and future ready manufacturing solutions

PSO Mapping

PSO Number	Name of the PSO	Justification
PSO1	Apply the knowledge of Artificial Intelligence to design, develop, and evaluate computational solutions for complex problems in diverse domains, such as healthcare, finance, and automation.	The project applies Artificial Intelligence, Computer Vision, and Vision-Language Models to develop an intelligent collaborative robotic system capable of perceiving its environment, understanding human instructions, and autonomously performing manufacturing tasks, thereby addressing complex automation challenges.
PSO2	Demonstrate expertise in using advanced ML tools, techniques, and frameworks to develop innovative solutions for data analysis, pattern recognition, and intelligent decision-making systems.	The project utilizes advanced machine learning frameworks, deep learning models, robotic middleware, and multimodal learning techniques to enable real-time perception, semantic reasoning, intelligent decision-making, and adaptive task execution in collaborative manufacturing environments.

Timelines



References

- [1] A. Brohan et al., "RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control," in *Proc. 7th Conf. on Robot Learning (CoRL)*, 2023, pp. 2165–2183.
- [2] M. J. Kim et al., "OpenVLA: An Open-Source Vision-Language-Action Model," in *Proc. 8th Conf. on Robot Learning (CoRL)*, vol. 270, 2025, pp. 2679–2713.
- [3] A. O'Neill et al., "Open X-Embodiment: Robotic Learning Datasets and RT-X Models," in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, Yokohama, Japan, 2024, pp. 6892–6903.
- [4] D. Guo et al., "What Matters in Building Vision-Language-Action Models for Generalist Robots," *Nature Machine Intelligence*, vol. 7, 2025.
- [5] F. Xia et al., "RoboCat: A Self-Improving Foundation Agent for Robotic Manipulation," *Transactions on Machine Learning Research (TMLR)*, 2024.
- [6] H. Liu, D. Guo, and A. Cangelosi, "Embodied Intelligence: A Synergy of Morphology, Action, Perception, and Learning," *ACM Computing Surveys*, vol. 57, no. 1, pp. 1–36, 2025.
- [7] H. Liu, Y. Zhu, K. Kato, A. Tsukahara, I. Kondo, and Y. Hasegawa, "Enhancing the LLM-Based Robot Manipulation Through Human–Robot Collaboration," *IEEE Robotics and Automation Letters*, vol. 9, pp. 6904–6911, 2024.
- [8] J. Fan and P. Zheng, "A Vision-Language-Guided Robotic Action Planning Approach for Ambiguity Mitigation in Human–Robot Collaborative Manufacturing," *Journal of Manufacturing Systems*, vol. 74, pp. 1009–1018, 2024.
- [9] J. Fan, Y. Yin, T. Wang, W. Dong, P. Zheng, and L. Wang, "Vision-Language Model-Based Human-Robot Collaboration for Smart Manufacturing: A State-of-the-Art Survey," *Frontiers of Engineering Management*, vol. 12, no. 1, pp. 177–200, 2025.

*Thank
you*

